

Conjugated equine estrogens and incidence of probable dementia and mild cognitive impairment in postmenopausal women: Women's Health Initiative Mem...

CONTEXT: The Women's Health Initiative Memory Study (WHIMS) previously found increased risk for dementia and no effect on mild cognitive impairment (MCI) in women treated with conjugated equine estrogens (CEE) plus medroxyprogesterone acetate (MPA). **OBJECTIVE:** To determine the effects of CEE alone and CEE plus MPA on incidence of probable dementia and MCI in older women. **DESIGN, SETTING, AND PARTICIPANTS:** Randomized, double-blind, placebo-controlled clinical trials of CEE (estrogen-alone trial) or CEE plus MPA (estrogen plus progestin trial) in community-dwelling women aged 65 to 79 years, conducted from June 1995 to July 8, 2002 (estrogen plus progestin; n = 4532), or to February 29, 2004 (estrogen-alone; n = 2947), in 39 of the 40 WHI clinical centers.

INTERVENTIONS: In the estrogen-alone trial, 1 daily tablet containing either 0.625 mg/d of CEE vs matching placebo; in the estrogen plus progestin trial, 1 daily tablet containing CEE (0.625 mg/d) plus MPA (2.5 mg/d) vs matching placebos. **MAIN OUTCOME MEASURES:** Probable dementia and MCI.

RESULTS: In the estrogen-alone trial, 47 participants were diagnosed with probable dementia, of whom 28 were assigned to CEE and 19 to placebo (hazard ratio [HR], 1.49; 95% confidence interval [CI], 0.83-2.66). Incidence rates for probable dementia in the estrogen-alone trial were statistically similar to those in the estrogen plus progestin trial (45 vs 22 per 10 000 person-years for CEE plus MPA vs placebo, respectively; P = .11). When data were pooled per the original WHIMS protocol, the overall HR for probable dementia was 1.76 (95% CI, 1.19-2.60; P = .005). After excluding participants with baseline Modified Mini-Mental State Examination scores at or below the screening cut point, the HR was 1.77 (95% CI, 0.74-4.23; P = .20) in the estrogen-alone trial and 2.19 (95% CI, 1.25-3.84; P = .006) in the pooled trials. In the estrogen-alone trial, 76 participants were diagnosed with MCI in the CEE group vs 58 in the placebo group (HR, 1.34; 95% CI, 0.95-1.89). In the combined trial data, the HR was similar (1.25; 95% CI, 0.97-1.60). In the estrogen-alone trial, 93 participants receiving CEE were diagnosed with either probable dementia or MCI vs 69 receiving placebo (HR, 1.38; 95% CI, 1.01-1.89; P = .04). **CONCLUSIONS:** Estrogen therapy alone did not reduce dementia or MCI incidence and increased the risk for both end points combined. Pooling data for estrogen alone and estrogen plus progestin resulted in increased risks for both end points. Use of hormone therapy to prevent dementia or cognitive decline in women 65 years of age or older is not recommended.